



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL
(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department	Computer Science and Engineering		
Course Name	Foundation of Computer Vision	Course Code	CSE 3172
Semester	II	Curriculum	2021
Name of the faculty	Dr SIDDALINGASWAMY P. C. Dr MURALI KRISHNA S. N.	Academic year	2023-24
No. of Contact Hours/Week	L T P C: 3 0 0 3		

COURSE OUTCOMES (CO'S)

At the end of this course, the student should be able to:		No. of Hours	Marks
CO1	Demonstrate the concepts of image formation and linear filtering.	10	25
CO2	Identify the mathematics behind feature detection and description methods.	9	25
CO3	Demonstrate the fundamental concepts in camera calibration.	6	20
CO4	Classify various object tracking algorithms.	6	20
CO5	Build object and scene recognition and categorization from images.	5	10
Total hours/ Marks		36	100

ASSESSMENT PLAN

Components	Quiz/Assignments	Midterm	End Semester/Make - Up Examinations
Duration	Quiz 1 [MCQ] Quiz 2 [Surprise] Quiz 3[Open book] [30 Mins each] Quiz 4[Take home]	Midterm [120 Mins each]	End sem exam [3 Hours]
Weightage	10 % [5M+5M] 10 % [5M+5M]	30% [15M+15M]	50% [50M]
Typology of Questions	Applying, Analysing and Evaluating	Understanding, Applying, Analysing and Evaluating	Understanding, Applying, Analysing and Evaluating
Pattern	Quizzes (MCQs and Descriptive)	Short Answer Questions, Descriptive Questions and Design Questions	Short Answer Questions, Descriptive Questions and Design Questions
Schedule	Quiz 1,2 3, 4: Calendared activity	Calendared activity	End of semester
Topics covered	Quiz 1: L1 to L8 Quiz 2: L9 to L18 Quiz 3: L21 to L30 Quiz 4: L31 to L36	L1 to L20	Exam: L1 to L36

LESSON PLAN

Lecture No.	Topic	CO's addressed
L0	Course overview	-
L1	Introduction to Computer Vision & Applications	CO1
L2	Image formation	CO1
L3	Image Sampling and Quantization	CO1
L4	Relationship between Pixels	CO1
L5	Image Enhancement in Spatial Domain	CO1
L6	Basic Intensity Transformations	CO1
L7	Histogram Equalization and Specification	CO1
L8	Spatial Filtering	CO1
L9	Image Segmentation – Edge Models (first and second order filters)	CO1
L10	Canny Edge Detection	CO1
L11	Hough Transform – Line Detection	CO2
L12	Otsu's Thresholding	CO2
L13	Morphological Dilation and Erosion	CO2
L14	Morphological Operations	CO2
L15	Mathematics of Harris Corner Detector	CO2
L16	Histogram of Gradients	CO2
L17	Difference of Gaussian for Multiscale features	CO2
L18	SIFT	CO2
L19	RANSAC	CO2
L20	Camera Model	CO3
L21	2D Transformations	CO3
L22	3D Transformations and Projections	CO3
L23	Computing Extrinsic Parameters	CO3

L24	Computing Intrinsic Parameters	CO3
L25	Stereo Correspondence	CO3
L26	Depth estimation	CO4
L27	Epipolar Geometry	CO4
L28	Fundament Matrix – 8 Point Algorithm	CO4
L29	Optical Flow	CO4
L30	Lucas Kanade Method	CO4
L31	KLT Algorithm	CO4
L32	SDL – Feature descriptors for object detection	CO5
L33	SDL – Colour based segmentation	CO5
L34	Case study – Skew correction in document	CO5
L35	Case study – Object localization	CO5
L36	Case study – Multiple object tracking	CO5

References:

References	
1	Rafael C. Gonzalez, Richard E. Woods, "Digital Image Processing", Third Edition, Pearson Education, 2012.
2	Richard Szeliski, Computer Vision: Algorithms and Applications, Springer 2011.
3	Milan Sonka, Vaclav Hlavac, Roger Boyle, "Digital Image Processing and Computer Vision", India Edition, Cengage Learning 2009.

Submitted by: P. C. SIDDALINGASWAMY

(Signature of the faculty)

Date: 31/07/2023

Approved by:

(Signature of HOD)

Date: 31/07/2023

COURSE PLAN – ADDITIONAL DETAILS

At the end of this course, the student should be able to:		No. of contact Hours	Marks	Program outcomes (PO's)	Learning outcomes (LO's)	PSO	BL
CO1	Demonstrate the concepts of image formation, and linear filtering.	10	25	1,5	1	–	2,3
CO2	Identify the mathematics behind feature detection and description methods.	9	25	1,3	3	–	2,3
CO3	Understand the fundamental concepts in camera calibration.	6	20	1,4	2	–	2,3
CO4	Classify various object tracking algorithms.	6	20	1,4,5	1,3	–	3,4
CO5	Build object and scene recognition and categorization from images.	5	10	1,3,4,5	1,3	1	5,6
Total hours/ Marks			100				

Course Articulation Matrix (NBA)

CO	PO1	PO2	PO3	PO4	PO5
CSE3172.1	3	–	–	–	1
CSE3172.2	1	–	3	–	–
CSE3172.3	2	–	-	2	–
CSE3172.4	1	–	-	4	1
CSE3172.5	2	-	2	2	2
Average Program Articulation Level	1.8	0	2.5	2.7	1.3

IET – Course Learning Outcomes (CLO's) mapping with AHEP LO's

Course Learning Outcome	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18
CSE3172.1	✓																	☐
CSE3172.2			✓															☐
CSE3172.3		✓																☐
CSE3172.4	✓																	☐
CSE3172.5	✓																	☐

Abbreviations

1. **CO – Course outcome**
2. **PO – Program outcome**
3. **PSO – Program Specific outcome**
4. **LO – Learning outcome**
5. **CLO – Course Learning outcome**
6. **BL – Blooms Taxonomy**
7. **AHEP – The Accreditation of Higher Education Programmes**